KOLEJ POLY-TECH MARA BATU PAHAT

DIPLOMA IN INFORMATION TECHNOLOGY (CT103)

COURSE WEB API DEVELOPMENT

(TSE3613)

ASSIGNMENT

PREPARED BY : TUAN AZEEM BIN TUAN AZIZ

STUDENT ID : BPJ241210438

PREPARED FOR : MADAM UMI KALSUM BT HASSAN

SUBMISSION DATE : 5TH FEBRUARY 2026

Integrating Social Media Data to Support Recruitment Using API-Based Data Exchange

---

1. INTRODUCTION

This assignment applies data conversion and interoperability principles to design a system that integrates social media data into a company's Human Resource (HR) recruitment platform. The system demonstrates how external platform data (LinkedIn, GitHub, Instagram, TikTok) can be retrieved through APIs, converted into standardized formats (JSON/XML), and exchanged using standard communication protocols such as REST over HTTPS.

The goal is to improve recruitment quality by enabling automated candidate evaluation based on professional activity, engagement behavior, and role relevance.

2. CASE STUDY ANALYSIS — INTEROPERABILITY CHALLENGES

2.1 Current Limitations of Existing HR Systems

Traditional HR systems mainly rely on resumes, application forms, and job portals. They do not provide automated access to candidates' social media activity. As a result:

- Recruiters must manually verify candidate portfolios
- Candidate engagement and online professionalism are difficult to assess
- Real-time professional activity is unavailable
- Hiring decisions may rely on incomplete information

These limitations reduce recruitment efficiency and decision accuracy.

2.2 Differences in Data Formats Across Platforms

Different social platforms return data in various formats:

- LinkedIn & GitHub → JSON
- Legacy or enterprise systems → XML
- Engagement metrics → Nested JSON structures

Challenges include inconsistent field naming, varied object nesting, and different data schemas.

Solution: A data transformation layer converts all external JSON/XML responses into a standardized internal HR data model.

2.3 Communication Protocols Used by Social Media APIs

Most modern social platforms use **RESTful APIs over HTTPS**. Some services also support **GraphQL** for structured queries.

Protocol	Usage	Benefit
REST	Standard HTTP requests	Simplicity, scalability
GraphQL	Flexible data querying	Efficient payloads
HTTPS	Secure transport	Data confidentiality

These protocols ensure interoperable and secure system communication.

2.4 Security and Privacy Considerations

Handling candidate data requires strict security and ethical compliance:

- OAuth 2.0 authentication
- HTTPS encrypted communication
- Role-based access control (RBAC)
- Candidate consent before data retrieval
- Secure API key and token storage

These measures ensure privacy protection and responsible data usage.

2.5 Integration Challenges

Challenge	Description	Mitigation
Data Mapping	Field mismatch	Schema mapping rules
Data Consistency	Conflicting records	Validation logic
Format Differences	JSON vs XML	Conversion module
API Rate Limits	Request restrictions	Throttling & caching
Policy Restriction	Platform rules	Compliance filters

3. SYSTEM PARADIGM SELECTION

Selected Paradigm: Microservices Architecture

3.1 Justification

Microservices architecture divides system functions into independent services. Each social media integration operates as a separate service, enabling scalability, fault isolation, and flexible maintenance.

This paradigm is suitable because social platforms frequently update APIs and require modular system updates.

3.2 Support for Modularity, Scalability, and Interoperability

- Each platform has an independent integration service
- Services scale separately based on API usage
- Standardized APIs ensure consistent communication
- Loose coupling prevents system-wide failures

3.3 Influence on API Design

Microservices architecture encourages:

- API Gateway for centralized routing
- Dedicated platform-specific microservices
- Unified internal JSON schema
- Middleware for data conversion and validation

3.4 Strengths and Limitations

Strengths

- High scalability
- Fault isolation
- Modular expansion
- Improved maintainability

Limitations

- Higher architectural complexity
- Increased network overhead
- Requires service monitoring and orchestration

4. API INTEGRATION PLAN

4.1 Key API Endpoints

LinkedIn API

GET /v2/me
GET /v2/experience
GET /v2/skills

GitHub API

GET /users/{username}/repos
GET /users/{username}/events

Instagram / TikTok API

GET /user/media
GET /user/engagement

4.2 API Request Example

GET https://api.linkedin.com/v2/me

Authorization: Bearer {access_token}

4.3 API Response Example (Raw JSON)


```
{
  "id": "abc123",
  "name": "John Doe",
  "skills": ["Java", "Cloud"],
  "connections": 450
}
```


4.4 Data Conversion & Transformation Process (Aligned with LAB Practice)

Step 1: Receive raw JSON/XML from external API

Step 2: Validate schema fields

Step 3: Rename and normalize fields

Step 4: Convert into standardized internal JSON format

Step 5: Store in HR database

Example Converted Output


```
{
  "candidateName": "John Doe",
  "skills": ["Java", "Cloud"],
  "networkSize": 450,
  "engagementScore": 78
}
```

4.5 Error Handling Strategy

Error Type	Handling Strategy
Timeout	Retry requests
Invalid Token	Refresh authentication
Missing Data	Log and skip
Rate Limit	Delay & queue

4.6 Request–Response Workflow

This section describes the end-to-end workflow of request and response between the HR system and external social media platforms, demonstrating how data exchange and interoperability are achieved using standard communication protocols.

1. **HR System Initiates Request**
   The HR system initiates a request to retrieve candidate-related data, such as professional history, skills, and engagement metrics. This request is sent using standard HTTP methods over a secure HTTPS connection.

2. **API Gateway Routes the Request**
   The API Gateway receives the incoming request and performs authentication and authorization checks. Based on the requested data sources, the gateway routes the request to the appropriate social media microservices. This routing mechanism abstracts the complexity of multiple external APIs from the HR system.

3. **Microservice Communicates with External API**
   Each microservice is responsible for interacting with a specific social media platform. The microservice constructs API requests according to the platform's specifications, including required parameters and OAuth 2.0 access tokens, and sends the request using REST or GraphQL protocols.

4. **External API Returns Response (JSON/XML)**
   The social media platforms return responses containing candidate data in either JSON or XML format. The responses may vary in structure, field naming, and completeness depending on the platform and access permissions.

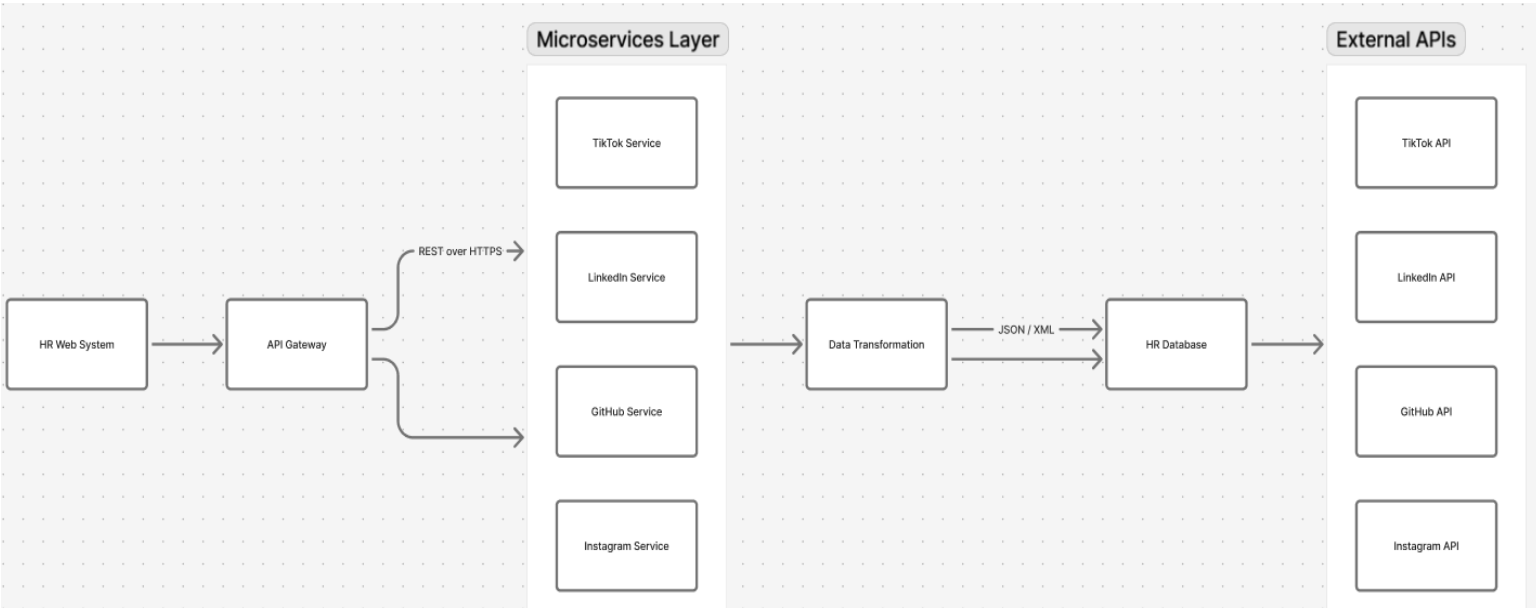
5. **Data Transformation and Normalization**
   The raw API responses are passed to the Data Transformation Layer, where the data is parsed and converted into a standardized internal format. This process includes schema mapping, format normalization, and validation to ensure consistency across all platforms.

6. **Data Storage in HR Database**
   The transformed and validated data is then stored in the HR database using a unified data model. Storing standardized data enables efficient querying, comparison, and long-term analysis while avoiding redundancy and inconsistency.

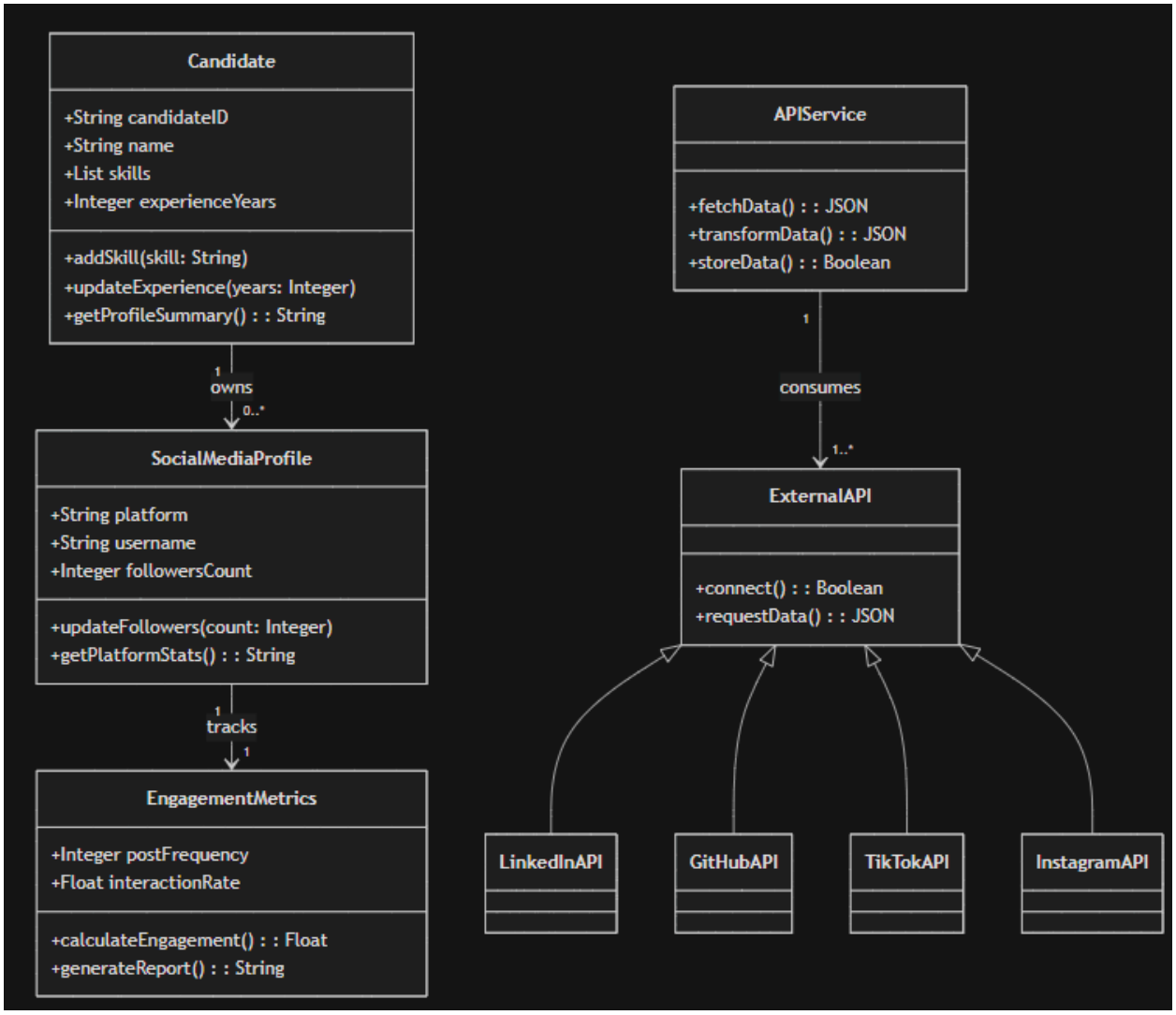
7. **HR Analytics and Candidate Evaluation**
   Finally, the HR analytics component processes the stored data to evaluate candidates. Metrics such as skill relevance, engagement level, and role alignment are analysed to support informed and data-driven recruitment decisions

5. SYSTEM DIAGRAMS

5.1 Architecture Diagram



5.2 UML Class Diagram (Structure)



6. RATIONALE & DISCUSSION

6.1 Why the Design is Effective

The microservices approach supports scalable API integration and ensures interoperable data exchange. It aligns with CLO 2 by demonstrating standardized communication protocols and structured data transformation.

6.2 Resolving Interoperability Challenges

Challenge	Solution
Format mismatch	Unified JSON schema
API differences	Dedicated services
Security	OAuth & HTTPS
Data inconsistency	Validation & filtering

6.3 How Data Conversion Improves HR Decisions

- Removes duplicate or inconsistent records
- Ensures consistent skill evaluation
- Aligns candidate profiles with job requirements
- Improves hiring accuracy

6.4 Benefits to Recruitment

- Faster candidate screening

- Improved hiring quality
- Reduced recruiter workload
- Data-driven hiring decisions

---

7. CONCLUSION

This assignment has examined the integration of social media data into a company's HR system to support more informed and data-driven recruitment decisions. By analysing the interoperability challenges associated with heterogeneous social media platforms such as LinkedIn, GitHub, Instagram, and TikTok, the study highlights the critical role of **data conversion and standardized communication protocols** in enabling effective system data exchange.

The case study demonstrates that traditional HR systems face significant limitations when accessing and processing external social media data due to fragmented data sources, inconsistent data formats, and varying API communication protocols. Differences in JSON and XML structures, schema variability, and platform-specific constraints require a well-defined integration approach to ensure data consistency and reliability.

To address these challenges, a **microservices-based system paradigm** was selected. This paradigm is particularly suitable for the recruitment case as it allows each social media platform to be integrated through independent services, reducing system coupling and improving maintainability. The use of REST and GraphQL APIs, combined with secure OAuth-based authentication, ensures interoperable communication between the HR system and external platforms while maintaining data security and privacy.

The inclusion of a dedicated **data conversion and transformation process** ensures that heterogeneous data retrieved from multiple APIs can be normalized into a unified internal HR data model. This process plays a crucial role in improving the accuracy of candidate evaluation by standardizing formats, resolving inconsistencies, and filtering irrelevant or incomplete data.

8. ADVANCED API RESPONSE EXAMPLES (TECHNICAL — LAB ALIGNED)

Example: GitHub API Raw Response

```
{
  "login": "devuser",
  "public_repos": 12,
  "followers": 89,
  "repos_url": "https://api.github.com/users/devuser/repos"
}
```

Transformed Internal HR Format

```
{
  "candidateUsername": "devuser",
  "portfolioStrength": 12,
  "networkScore": 89,
  "platform": "GitHub"
}
```

Example: Engagement Scoring Formula

EngagementScore = (PostFrequency × 0.4) + (Interactions × 0.6)

10. FINAL RATIONALE

The proposed system enables recruiters to make decisions based on verified digital professional activity rather than relying only on resumes. By converting inconsistent API data into a unified internal format, the system ensures fairness, accuracy, and reliability in candidate evaluation.

Microservices architecture improves adaptability, allowing the organization to integrate new platforms in the future without redesigning the entire system. The use of REST APIs, JSON/XML transformation, and secure authentication mechanisms demonstrates strong application of CLO 2 principles.

Overall, the design strengthens recruitment efficiency, enhances data-driven hiring, and demonstrates effective real-world system interoperability.

11. REFERENCES

LinkedIn Developer Documentation. (2024) : https://developer.linkedin.com/
GitHub REST API Documentation. (2024) : https://share.google/CNSkQ5YWbTTj5fN7O
Fielding, R. (2000). Architectural Styles and REST : https://share.google/lCLy6174fkT9EsUa3
ISO/IEC 25010 Software Quality Standards. (2022) : https://share.google/aJXmH8ISfVartS3kT